

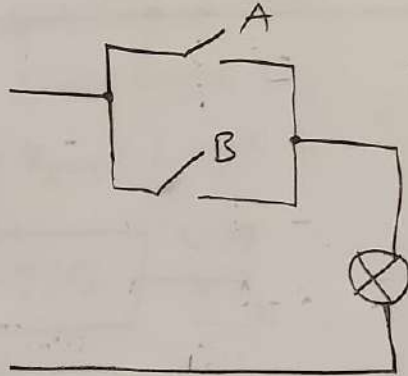
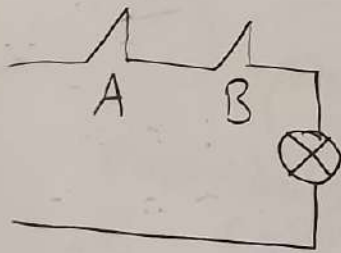
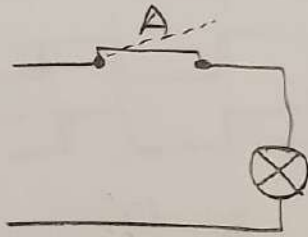
LAWS OF BOOLEAN ALGEBRA

Operators

NOT (—,)

AND (.)

ORC (+)



NOT operator (Complementation)

X	F = $\bar{X}$
0	1
1	0



$\bar{\bar{X}} = X \rightarrow$ Involution

AND operator

Dec	x	y	F = $x \cdot y$	AND LAWS
0	0	0	0	$x \cdot 0 = 0$ (Absorber)
1	0	1	0	$x \cdot 1 = x$ (Neutral)
2	1	0	0	$x \cdot \bar{x} = 0$
3	1	1	1	$x \cdot x = x$

OR Operator

Dec	x	y	F = $x + y$	OR LAWS
0	0	0	0	$0 + x = x$
1	0	1	1	$1 + x = 1$
2	1	0	1	$x + \bar{x} = 1$
3	1	1	1	$x + x = x$

$x \cdot x = x$

Concept of Duality

AND ve OR eşitsizlikleri birbirlerinin dualidir.

$$0 \Longleftrightarrow 1$$

$$(\cdot) \Longleftrightarrow (+)$$

$$X+1=1$$

- Associative Laws (Birleşme Yasası) $\overline{\overline{X}} = X$
- Commutative Laws (Değişme Yasası)
- Distributive Laws (Dağıtma Yasası)
- Absorptive Laws (Yutma Yasası)

Associative Laws

$\forall x, y, z \in B$ (Boole set)

$$(x \cdot y) \cdot z = x \cdot (y \cdot z) = x \cdot y \cdot z$$

$$(x + y) + z = x + (y + z) = x + y + z$$

Commutative Laws

$$x \cdot y \cdot z = y \cdot x \cdot z = y \cdot z \cdot x$$

$$x + y + z = y + x + z = y + z + x$$

Distributive Laws

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

$$x + (y \cdot z) = (x + y) \cdot (x + z)$$

Absorptive Laws

$$x \cdot (\bar{x} + y) = x \cdot y$$

$$x + (\bar{x} \cdot y) = x + y$$

De Morgan Laws

$$\overline{x \cdot y} = \bar{x} + \bar{y}$$

$$\overline{x + y} = \bar{x} \cdot \bar{y}$$

Complementation

$$x + \bar{x} = 1$$

$$x \cdot \bar{x} = 0$$

Neutr Element

$$x + 0 = x$$

$$x \cdot 1 = x$$

Absorption Element

$$x \cdot 0 = 0$$

$$x + 1 = 1$$

Involution

$$\overline{\overline{x}} = x$$

Idempotents

$$x \cdot x = x$$

$$x + x = x$$



Exclusive OR (EXOR, XOR) $\oplus$ $\oplus$

Equivalente = Exclusive NOR $\odot$ $\odot$

Dec	x	y	$x \oplus y$	$x \odot y$
0	0	0	0	1
1	0	1	1	0
2	1	0	1	0
3	1	1	0	1

EXOR

$$x \oplus 0 = x$$

$$x \oplus 1 = \bar{x}$$

$$x \oplus x = 0$$

$$x \oplus \bar{x} = 1$$

EX NOR

$$x \odot 1 = x$$

$$x \odot 0 = \bar{x}$$

$$x \odot x = 1$$

$$x \odot \bar{x} = 0$$

$$x \odot x = 1$$

$$x \oplus y = x \cdot \bar{y} + \bar{x} \cdot y = \overline{x \odot y}$$

$$\bar{x} \cdot y + x \cdot \bar{y}$$

$$x \odot y = x \cdot y + \bar{x} \cdot \bar{y} = \overline{x \oplus y}$$

Associative Laws

$$(x \odot y) \odot z = x \odot (y \odot z)$$

$$(x \oplus y) \oplus z = x \oplus (y \oplus z)$$

Commutative Laws

$$x \odot y \odot z = x \odot z \odot y = z \odot x \odot y$$

$$x \oplus y \oplus z = x \oplus z \oplus y = z \oplus x \oplus y$$

Absorptive Laws

$$x \cdot (\bar{x} \oplus y) = x \cdot y$$

$$x + (\bar{x} \odot y) = x + y$$

Distributive Laws

$$x \cdot (y \oplus z) = (x \cdot y) \oplus (x \cdot z)$$

$$x + (y \odot z) = (x + y) \odot (x + z)$$

De Morgan Laws

$$\overline{x \oplus y} = \bar{x} \odot \bar{y} = x \odot y$$

$$\overline{x \odot y} = \bar{x} \oplus \bar{y} = x \oplus y$$

Positive Logic

Negative Logic

$$HV \longleftrightarrow 1(H) = 0(L)$$

$$LV \longleftrightarrow 0(H) = 1(L)$$

Logic Gates

NOT Gate

AND Gate

OR Gate

NOR Gate

NAND Gate

EXOR Gate

EQV Gate

NOT GATE = INVERTER

X	F
0	1
1	0



X	Y	$x \oplus y$
0	0	0
0	1	1
1	0	1
1	1	0

Active High to Active Low

X	F = $\bar{X}$
LV	HV
HV	LV

Active Low to Active High

X(H)	X(L)
0	0
1	1

Active Low Indicator (Dubble)

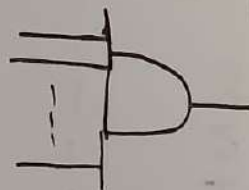
X(L)	X(H)
1	1
0	0

$$\overline{(x \cdot \bar{x})} \oplus (x \cdot y) = x \cdot y$$



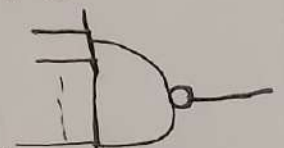
AND Gate

Dec	x	y	F = x · y
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1



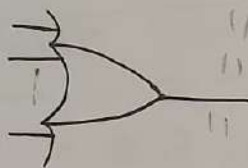
NAND Gate

Dec	x	y	x · y	F = $\overline{x \cdot y}$
0	0	0	0	1
1	0	1	0	1
2	1	0	0	1
3	1	1	1	0



OR Gate

Dec	x	y	$F = x + y$
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1



NOR Gate

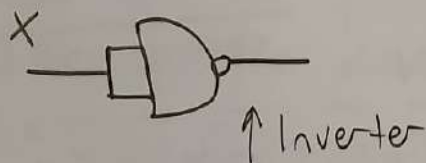
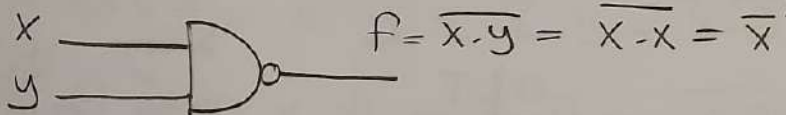
Dec	x	y	$x + y$	$F = \overline{x + y}$
0	0	0	0	1
1	0	1	1	0
2	1	0	1	0
3	1	1	1	0



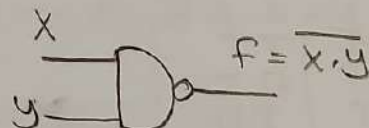
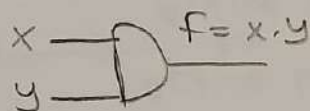
UNIVERSALITY OF NAND GATES AND NOR GATES

NAND Gate Implementation

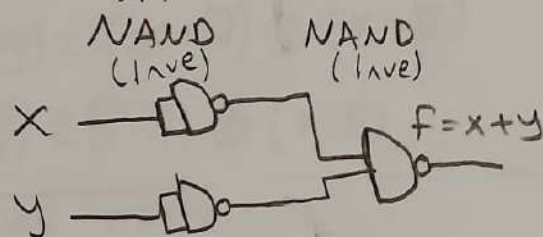
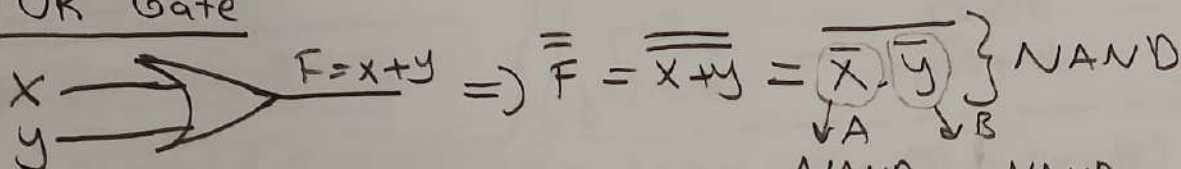
• Inverter



AND Gate

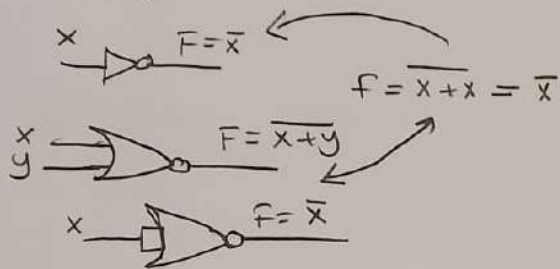


OR Gate

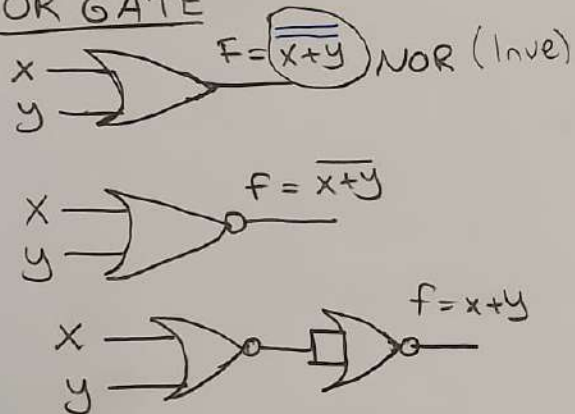


NOR GATE Implementation

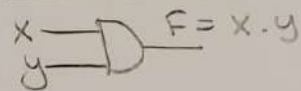
• Inverter



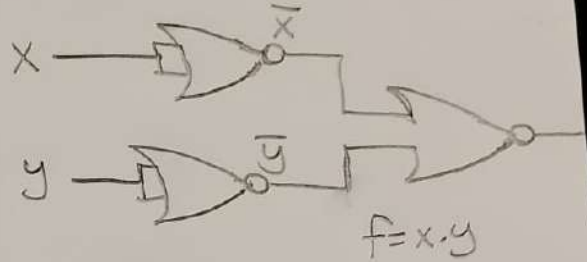
• OR GATE



AND GATE



$$F = \overline{x+y} = \overline{\overline{\overline{x}} + \overline{\overline{y}}} \quad \left. \begin{array}{l} \text{NOR A} \quad \text{NOR B} \end{array} \right\} \text{NOR}$$



2. hafta

